

JavaScript Strings

What is a String?

Definition

A String is a sequence of characters used to represent text in JavaScript.

A string can contain:

Letters

Numbers

Symbols

Spaces

Special Characters

Strings are enclosed in:

Single Quotes (' ')

Double Quotes (" ")

Backticks (```)

Why Do We Use Strings?

Strings are used to store textual data such as:

User Name

Email

Password

Address

Messages

Product Name

City Name

Creating Strings

There are three ways to create a string.

1. Using Double Quotes

```
let name = "Rokon";
```

```
console.log(name);
```

Output

Rokon

2. Using Single Quotes

```
let country = 'Bangladesh';
```

```
console.log(country);
```

Output

Bangladesh

3. Using Backticks (Template Literals)

```
let language = JavaScript ;
```

```
console.log(language);
```

Output

JavaScript

String Examples

```
let firstName = "MD"; let lastName = "Rokonuzzaman";
```

```
console.log(firstName); console.log(lastName);
```

Output

MD Rokonuzzaman

String Length

Definition

The length property returns the total number of characters in a string.

It counts:

Letters

Numbers

Spaces

Symbols

Everything counts as one character.

Syntax

```
string.length
```

Example 1

```
let name = "JavaScript";  
console.log(name.length);
```

Output

10

Example 2

```
let text = "Hello World";  
console.log(text.length);
```

Output

11

Explanation

H e l l o _ W o r l d 1 2 3 4 5 6 7 8 9 10 11

Space is also counted.

Real Project Example

Check password length.

```
let password = "abc12345";  
if(password.length >= 8){ console.log("Strong Password"); }else{ console.log("Weak Password"); }
```

Output

Strong Password

Character Access

Definition

Each character in a string has its own position called an index.

JavaScript starts counting from 0.

Example

```
let language = "JavaScript";
```

Character

J a v a S c r i p t

Index

0 1 2 3 4 5 6 7 8 9

Access Using Index

```
let language = "JavaScript";
```

```
console.log(language[0]); console.log(language[1]); console.log(language[4]);
```

Output

J a S

Last Character

```
let language = "JavaScript";
```

```
console.log(language[language.length-1]);
```

Output

t

Loop Through a String

```
let text = "Code";
```

```
for(let i=0;i<text.length;i++){
```

```
    console.log(text[i]);
```

```
}
```

Output

C o d e

Immutable

Definition

Strings are immutable.

Immutable means once a string is created, its characters cannot be changed directly.

You cannot modify an existing character using its index.

Wrong Example

```
let name = "Rokon";
```

```
name[0] = "M";
```

```
console.log(name);
```

Output

Rokon

Nothing changes.

Why?

JavaScript does not allow changing individual characters of a string.

Instead, create a new string.

Correct Way

```
let name = "Rokon";  
  
name = "Mokon";  
  
console.log(name);
```

Output

Mokon

Another Example

```
let text = "Hello";  
  
text = text + " World";  
  
console.log(text);
```

Output

Hello World

JavaScript creates a new string instead of changing the old one.

Escape Characters

Definition

Escape characters allow special characters to be included inside a string.

They start with a backslash ().

Common Escape Characters

Escape	Meaning
--------	---------

" Double Quote	' Single Quote	\ Backslash	\n New Line	\t Tab
------------------	------------------	---------------	---------------	----------

Double Quote Example

```
let text = "He said, "Hello"";  
  
console.log(text);
```

Output

He said, "Hello"

Single Quote Example

```
let text = 'It's JavaScript';
```

```
console.log(text);
```

Output

It's JavaScript

New Line

```
let text = "HTML\nCSS\nJavaScript";
```

```
console.log(text);
```

Output

HTML CSS JavaScript

Tab

```
let text = "Name\tAge";
```

```
console.log(text);
```

Output

Name Age

Backslash

```
let path = "C:\\Users\\Admin";
```

```
console.log(path);
```

Output

C:\Users\Admin

Best Practices

✅ Use double quotes or single quotes consistently.

```
let name = "Rokon";
```

✅ Use template literals when inserting variables.

```
let name = "Rokon";
```

```
console.log( Welcome ${name} );
```

✅ Use .length instead of manually counting characters.

Common Mistakes

Forgetting Quotes



```
let name = Rokon;
```

Error

Wrong Index

```
let name = "Java";
```

```
console.log(name[10]);
```

Output

undefined

Trying to Modify a Character



```
let name = "Java";
```

```
name[0] = "C";
```

Output

Java

Interview Questions

What is a String?

A string is a sequence of characters used to represent text in JavaScript.

What is String Length?

The length property returns the total number of characters in a string.

What does Immutable mean?

Immutable means the value cannot be changed after it is created. Strings in JavaScript are immutable.

What is Character Index?

An index is the position of a character inside a string. Indexing starts from 0.

Summary

Topic	Description
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String	Sequence of characters	Quotes	Single, Double, Backticks	length	Returns total characters
Character Access	Uses index starting from 0	Immutable	Characters cannot be		

changed directly || Escape Characters | Specialcharacters inside strings |

JavaScript String Methods (Part 2)

This section covers the most commonly used string methods in JavaScript.

`toLowerCase()`

`toUpperCase()`

`trim()`

`slice()`

`substring()`

`concat()`

`includes()`

`startsWith()`

`endsWith()`

1. `toLowerCase()`

Definition

The `toLowerCase()` method converts all characters in a string to lowercase.

It does not change the original string. It returns a new string.

Syntax

```
string.toLowerCase()
```

Example

```
let text = "HELLO WORLD";  
console.log(text.toLowerCase());
```

Output

hello world

Real Project Example

Email comparison.

```
let email = "Admin@Gmail.com";  
console.log(email.toLowerCase());
```

Output

admin@gmail.com

2. toUpperCase()

Definition

The toUpperCase() method converts all characters to uppercase.

It returns a new string.

Syntax

```
string.toUpperCase()
```

Example

```
let text = "javascript";  
console.log(text.toUpperCase());
```

Output

JAVASCRIPT

Real Project Example

```
let country = "bangladesh";  
console.log(country.toUpperCase());
```

Output

BANGLADESH

3. trim()

Definition

The trim() method removes whitespace from the beginning and the end of a string.

It does not remove spaces between words.

Syntax

```
string.trim()
```

Example

```
let text = " Hello World ";  
console.log(text.trim());
```

Output

Hello World

Without trim()

```
" Hello World "
```

After trim()

```
"Hello World"
```

Real Project Example

User input.

```
let username = " Rokon ";
```

```
console.log(username.trim());
```

Output

```
Rokon
```

4. slice()

Definition

The slice() method extracts a part of a string.

The original string is not changed.

Syntax

```
string.slice(start, end)
```

Start index is included.

End index is excluded.

Example

```
let language = "JavaScript";
```

```
console.log(language.slice(0,4));
```

Output

```
Java
```

Example 2

```
console.log(language.slice(4));
```

Output

Script

Negative Index

```
let text = "JavaScript";  
console.log(text.slice(-6));
```

Output

Script

Real Project Example

```
let phone = "01712345678";  
console.log(phone.slice(-4));
```

Output

5678

5. substring()

Definition

The substring() method extracts characters from a string.

Unlike slice(), it does not accept negative indexes.

Syntax

```
string.substring(start,end)
```

Example

```
let language = "JavaScript";  
console.log(language.substring(0,4));
```

Output

Java

Example

```
console.log(language.substring(4));
```

Output

Script

Difference Between slice() and substring()

slice()	substring()
Supports negative index	Does not support negative index
More commonly used	Less commonly used

6. concat()

Definition

The concat() method joins two or more strings.

Syntax

```
string.concat(string2)
```

Example

```
let firstName = "MD"; let lastName = "Rokon";  
console.log(firstName.concat(" ",lastName));
```

Output

MD Rokon

Another Way

```
console.log(firstName + " " + lastName);
```

Modern JavaScript usually uses:

```
${firstName} ${lastName}
```

7. includes()

Definition

The includes() method checks whether a string contains a specific value.

It returns:

true

false

Syntax

```
string.includes(searchValue)
```

Example

```
let language = "JavaScript";  
console.log(language.includes("Script"));
```

Output

true

Example

```
console.log(language.includes("Python"));
```

Output

false

Real Project Example

```
let email = "admin@gmail.com";
```

```
console.log(email.includes("@"));
```

Output

true

8. startsWith()

Definition

The startsWith() method checks whether a string starts with specific characters.

Returns:

true

false

Syntax

```
string.startsWith(value)
```

Example

```
let language = "JavaScript";
```

```
console.log(language.startsWith("Java"));
```

Output

true

Example

```
console.log(language.startsWith("Script"));
```

Output

false

Real Project Example

Check URL.

```
let url = "https://google.com";
```

```
console.log(url.startsWith("https"));
```

Output

true

9. endsWith()

Definition

The endsWith() method checks whether a string ends with specific characters.

Returns:

true

false

Syntax

```
string.endsWith(value)
```

Example

```
let language = "JavaScript";  
console.log(language.endsWith("Script"));
```

Output

true

Example

```
console.log(language.endsWith("Java"));
```

Output

false

Real Project Example

Check file extension.

```
let file = "resume.pdf";  
console.log(file.endsWith(".pdf"));
```

Output

true

Summary Table

Method	Purpose	Returns
--------	---------	---------

||-| | toLowerCase() | Converts to lowercase | New String | |toUpperCase() | Converts to uppercase | New String | | trim() |Removes leading and trailing spaces | New String | | slice() |Extracts part of a string | New String | | substring() | Extractspart of a string | New String | | concat() | Joins strings | NewString | | includes() | Checks if text exists | Boolean | |startsWith() | Checks starting text | Boolean | | | endsWith() |Checks ending text | Boolean |

Most Frequently Used Methods



trim()

toLowerCase()

includes()

slice()



startsWith()

endsWith()

concat()



substring()

Interview Questions

Q1. What is the difference between slice() and substring()?

slice() substring()

Supports negative index Does not support negative indexMore flexible Simpler

Q2. Which method removes extra spaces?

trim()

Q3. Which method checks whether a string contains another string?

includes()

Q4. Which method converts a string to lowercase?

toLowerCase()

Q5. Which method checks the beginning of a string?

startsWith()

Q6. Which method checks the ending of a string?

endsWith()